Learning Objective 2.6: I can measure the impedance and S-parameters of an antenna using a vector network analyzer and interpret the results.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Four names for one number.** A one-port measurement gets reported four different ways depending on which menu you open. Reference impedance is $Z_0 = 50\,\Omega$ throughout.

   (a) An antenna reads $|S_{11}| = -14$ dB at its design frequency. Find $|\Gamma|$, the VSWR, and the percentage of incident power reflected.

   $|\Gamma|$ = 0.199

   VSWR = 1.50

   P_{refl} = 4.0%

   $$|\Gamma| = 10^{-14/20} = 0.199$$
   $$\text{VSWR} = \frac{1+0.199}{1-0.199} = \frac{1.199}{0.801} = 1.50$$
   $$\frac{P_{refl}}{P_{inc}} = |\Gamma|^2 = 0.0398 \rightarrow 4.0\%$$

   (b) A second antenna shows VSWR $= 2.0$ at the same frequency. What is its $|S_{11}|$ in dB?

   $|S_{11}|$ = -9.5 dB

   $$|\Gamma| = \frac{\text{VSWR}-1}{\text{VSWR}+1} = \frac{1}{3} = 0.333$$
   $$|S_{11}| = 20\log_{10}(0.333) = -9.5 \text{ dB}$$

   It misses the -10 dB specification, but only just.

   (c) Why is -10 dB the conventional acceptance level for an antenna match? Answer in terms of power, in one or two sentences.

   At -10 dB, $|\Gamma|^2 = 0.10$: exactly 10% of the delivered power comes back and 90% goes into the antenna. Below that level the reflected power is a small correction to the link budget, so making the match better buys very little, while making it worse costs quickly.

(d) Lesson 7 predicted $Z_{\text{in}} = 73 + j42.5\,\Omega$ for an ideal half-wave dipole. Compute $|\Gamma|$, $|S_{11}|$ in dB, and the VSWR. Does the textbook dipole meet the -10 dB specification, and what one change fixes it?

$$\Gamma = \frac{23 + j42.5}{123 + j42.5}$$
$$|\Gamma| = \frac{48.3}{130.1} = 0.371$$
$$|S_{11}| = 20\log_{10}(0.371) = -8.6\text{ dB}$$
$$\text{VSWR} = \frac{1.371}{0.629} = 2.18$$

It **fails**. The fix is to shorten the element to about 0.47–0.48λ so the reactance goes to zero and R_{in} falls to roughly $70\,\Omega$:

$$|\Gamma| = \frac{20}{120} = 0.167 \rightarrow -15.6\text{ dB}, \qquad \text{VSWR} = 1.40$$

The $42.5\,\Omega$ of reactance is the entire difference between a marginal antenna and a good one.

$|S_{11}|$ = -8.6 dB

VSWR = 2.18

Passes? no

2. **From a marker readout to an impedance.** A calibrated VNA marker sitting at 1.575 GHz on a small GPS-band element reads $S_{11} = 0.33\,\angle 155°$.

   (a) Write Γ in rectangular form and find the input impedance Z_{in}.

   Z_{in} = $\boxed{26.1 + j8.2\,\Omega}$

$$\Gamma = 0.33\,(\cos 155° + j\sin 155°) = -0.299 + j0.140$$

$$Z_{\text{in}} = Z_0\,\frac{1+\Gamma}{1-\Gamma} = 50\,\frac{0.701 + j0.140}{1.299 - j0.140}$$

   Multiplying numerator and denominator by the conjugate of the denominator,

$$\begin{aligned} Z_{\text{in}} &= 50\,\frac{0.891 + j0.279}{1.707} \\ &= 50\,(0.522 + j0.163) \\ &= 26.1 + j8.2\,\Omega \end{aligned}$$

   (b) The reactance came out positive. Is the element electrically long or short at 1.575 GHz, is its resonance above or below that frequency, and which way would you trim it?

   Positive reactance is inductive, which is what a wire antenna looks like when it is **electrically long**. Its resonance ($X = 0$) therefore lies **below** 1.575 GHz. To move the resonance up onto the design frequency you trim the element **shorter**.

   (c) Find $|S_{11}|$ in dB and the VSWR. Does this element meet the -10 dB specification as measured?

   $|S_{11}|$ = $\boxed{-9.6 \text{ dB}}$

   VSWR = $\boxed{1.99}$

$$|S_{11}| = 20\log_{10}(0.33) = -9.6 \text{ dB}$$

$$\text{VSWR} = \frac{1.33}{0.67} = 1.99$$

   No – it misses the specification by 0.4 dB .

   (d) Suppose you trim the element perfectly, so at 1.575 GHz the reactance is zero and the resistance stays near $26\,\Omega$. Will the match pass then? Compute it, and state what the result tells you about relying on trimming alone.

$$\Gamma = \frac{26-50}{26+50} = -0.316 \quad\rightarrow\quad |S_{11}| = -10.0 \text{ dB}, \qquad \text{VSWR} = 1.92$$

   It lands *exactly* on the specification with no margin at all. Trimming only removes reactance; it cannot move the resistance. An element whose radiation resistance sits far from $50\,\Omega$ needs a matching network or a different feed arrangement, not a shorter piece of wire.

3. **Bandwidth off a sweep.** A calibrated, load-verified sweep of a wire dipole produced the following magnitudes.

f (MHz)	850	870	880	890	900	905	910	920	930	950
$\lvert S_{11}\rvert$ (dB)	-3.8	-6.5	-8.8	-12.4	-17.7	-19.4	-17.8	-12.5	-9.0	-5.2

(a) Identify the resonant frequency and say how you know.

The deepest point of the sweep is -19.4 dB at 905 MHz, and the trace is symmetric about it. That is the resonance: the reactance passes through zero there, so the reflection is smallest.

f_0 = 905 MHz

(b) Interpolate the two -10 dB crossings and report the impedance bandwidth both in MHz and as a percentage.

Lower crossing, between 880 MHz (-8.8 dB) and 890 MHz (-12.4 dB):

$$f_1 = 880 + 10\,\frac{10 - 8.8}{12.4 - 8.8} = 883 \text{ MHz}$$

Upper crossing, between 920 MHz (-12.5 dB) and 930 MHz (-9.0 dB):

$$f_2 = 920 + 10\,\frac{12.5 - 10}{12.5 - 9.0} = 927 \text{ MHz}$$

$$\text{BW} = 927 - 883 = 44 \text{ MHz}, \qquad \frac{44}{905} = 4.8\%$$

BW = 44 MHz

Fractional = 4.8%

(c) At resonance the Smith-chart marker sits on the real axis, to the *right* of center. Find Z_{in} at resonance.

On the real axis Γ is purely real, and right of center means it is positive, so $R_{\text{in}} > 50\,\Omega$:

$$|\Gamma| = 10^{-19.4/20} = 0.107$$
$$Z_{\text{in}} = 50\,\frac{1 + 0.107}{1 - 0.107} = 50\,(1.240) = 62\,\Omega$$

Without the Smith chart you could not choose between $62\,\Omega$ and $40\,\Omega$: the magnitude alone does not carry the sign.

Z_{in} = $62 + j0\,\Omega$

(d) A thin wire dipole typically shows 3–10% impedance bandwidth. Does this element fit the rule of thumb, and what physical change would widen it?

Yes: 4.8% sits in the lower half of the usual 3–10% range, which is what a thin conductor gives. Bandwidth widens when the antenna's Q drops, so use a fatter conductor – a thicker rod, a sleeve, a cage of wires, or a biconical element – which lowers the stored-to-radiated energy ratio.

4. **Calibration and the reference plane.** A classmate calibrates short–open–load at the end of the test cable, then screws a 10 cm RG-58 pigtail ($v_f = 0.66$) between that connector and the antenna before sweeping.

   (a) Name the three error terms a one-port SOL calibration removes, one sentence each, and explain why exactly three standards are required.

   Directivity: the coupler that should sample only the returning wave leaks some of the outgoing wave into the same receiver, so the VNA sees a reflection even from a perfect load. **Source match**: the test port is not exactly $50\,\Omega$, so energy returned by the antenna is partly re-reflected back at it. **Tracking**: the reference and test receiver paths have different gain and phase, and both drift with frequency. Three unknowns require three independent known loads, which is exactly what short ($\Gamma = -1$), open ($\Gamma = +1$), and load ($\Gamma = 0$) provide.

   (b) Describe what the extra pigtail does to the measured $|S_{11}|$ magnitude and, separately, to the measured impedance. Why is this combination dangerous?

   A short low-loss pigtail changes the *magnitude* hardly at all, so the dB plot still looks correct and the resonant frequency still reads correctly. It adds round-trip *phase*, which rotates the entire trace around the Smith chart, so the impedance readout is wrong – possibly wildly wrong, including the sign of the reactance. That combination is dangerous because the plot that still looks correct is the one most likely to be trusted.

   (c) At 915 MHz, find the guided wavelength in the pigtail, its length in wavelengths, and the round-trip phase shift $2\beta\ell$ it adds.

   λ_g = $\boxed{21.6\text{ cm}}$

   $2\beta\ell$ = $\boxed{333^\circ}$

$$\lambda_g = \frac{c\,v_f}{f} = \frac{(3\times10^8)\,(0.66)}{915\times10^6} = 0.216\text{ m} = 21.6\text{ cm}$$

$$\frac{\ell}{\lambda_g} = \frac{10}{21.6} = 0.462$$

$$2\beta\ell = 2\,(360^\circ)\,(0.462) = 333^\circ$$

   That is nearly a full rotation of the chart.

   (d) At what frequency is this pigtail exactly a half guided wavelength long? Why does that frequency matter to your classmate?

   f = $\boxed{990\text{ MHz}}$

$$\lambda_g = 2\ell = 0.20\text{ m}$$

$$f = \frac{c\,v_f}{\lambda_g} = \frac{(3\times10^8)\,(0.66)}{0.20} = 990\text{ MHz}$$

   A half-wavelength line repeats its load impedance, so at 990 MHz (and multiples of it) the un-de-embedded reading happens to be correct. At every other frequency it is rotated. Obtaining the correct answer at that one frequency is a coincidence rather than a calibration; fix it with port extension or by calibrating at the antenna connector.

5. **What the VNA cannot see.** The instrument reports one complex number per frequency. This problem is about the questions that number cannot answer.

   (a) You solder a $50\,\Omega$ chip resistor across the connector and sweep it. Predict $|S_{11}|$, the VSWR, and the radiated power. What does the result prove about what S_{11} measures?

   $\Gamma = 0$, so $|S_{11}|$ drops to the noise floor of the instrument and the VSWR reads 1.00 at every frequency: a perfect match. It radiates **nothing** – all of the delivered power becomes heat. S_{11} therefore measures **mismatch only**. It cannot separate power that left as radiation from power that died as loss, so it can never confirm efficiency.

   (b) Antenna A measures -25 dB with 2.0 dBi of gain; antenna B measures -9 dB with 5.0 dBi. Find the mismatch loss of each and decide which one puts more power into its main beam.

   L_A = 0.01 dB

   L_B = 0.58 dB

   Better: B

$$
\begin{aligned}
|\Gamma_A| &= 10^{-25/20} = 0.056, \qquad |\Gamma_A|^2 = 0.0032\\
L_A &= -10\log_{10}(1-0.0032) = 0.01 \text{ dB}\\
|\Gamma_B| &= 10^{-9/20} = 0.355, \qquad |\Gamma_B|^2 = 0.126\\
L_B &= -10\log_{10}(1-0.126) = 0.58 \text{ dB}
\end{aligned}
$$

   Realized gains are $2.0 - 0.01 = 2.0$ dBi and $5.0 - 0.58 = 4.4$ dBi. **Antenna B wins by about 2.4 dB**, despite its much worse S_{11}. Mismatch loss is small until $|S_{11}|$ gets much worse than -10 dB.

   (c) During the lab you bring a hand within a few centimetres of the element and the dip moves. State which way the resonant frequency shifts, what happens to the depth of the dip, and why neither effect is instrument error.

   The resonance shifts **down** and the dip usually gets **shallower and broader**. Tissue is a lossy dielectric: placed in the reactive near field it adds capacitive loading, which lowers f_0, and it adds loss, which lowers the antenna's Q. The input impedance changed, so the VNA is reporting a correct measurement – the hand became part of the antenna.

   (d) Your antenna measures -30 dB at the design frequency, yet the link will not close. Name two things a VNA cannot see that would explain this, and say how you would check one of them.

   (1) **Efficiency**: a lossy structure – corroded joints, a resistive loading element, wet material in the near field – absorbs the power it accepts while still presenting a good match. (2) **Pattern or polarization**: the beam may point somewhere other than the far end of the link, or be cross-polarized to it. Check the first with a gain comparison against a standard-gain antenna on the range (Lesson 11): if the measured gain falls well short of the directivity you predicted, the missing decibels are loss.

Documentation: ____________________